

Evaluation of the Digital Accessibility of Institutional Portals and Electronic Ombudsman Channels in the Municipalities of Mato Grosso

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Abstract. Research Context: Digital accessibility is a legal requirement for Brazilian government portals, including municipal ombudsman services. **Scientific and/or Practical Problem:** There is no systematic evidence on the accessibility of the portals of Mato Grosso's city halls and city councils, nor on whether it is maintained in ombudsman channels, often hosted separately. **Proposed Solution and/or Analysis:** The study covered all 142 municipalities of the state and evaluated three environments (city hall portal, city council portal and ombudsman channel) with two automated WCAG 2.1-based tools, AMAWeb and AccessMonitor. **Related IS Theory:** The study draws on the DeLone & McLean IS Success Model, in which system quality, operationalized here as compliance with accessibility guidelines, is an antecedent of use and user satisfaction. **Research Method:** A descriptive, comparative and quantitative census study. Portal addresses came from the Radar da Transparência Pública; ombudsman channels were located manually and classified by type. Scores were correlated with population, GDP per capita and HDI (Spearman). **Summary of Results:** The three environments were evaluated in 88 municipalities. Mean scores were close (AMAWeb: 7.65 city halls, 7.68 councils, 7.98 ombudsman; AccessMonitor: 7.01, 6.89 and 7.48), but within the same municipality the mean absolute difference between city hall and ombudsman was 0.92 and 1.17 points, and between city hall and council 0.98 and 1.19, so one environment does not represent another. Outsourced ombudsman solutions outscored their portals in most municipalities. Insufficient contrast was the most frequent error, and ombudsman channels concentrated indicators of keyboard and focus barriers. Correlations with socioeconomic indicators were weak. **Contributions and Impact to IS area:** The work offers an unprecedented diagnosis of municipal digital accessibility covering the whole state of Mato Grosso, shows that each channel must be evaluated separately, and provides a replicable procedure and a baseline for monitoring digital government in the state.

1. Introduction

Digital accessibility is a condition for all people, including people with disabilities, to be able to access information and public services offered over the internet. In Brazil, this right is ensured by the Brazilian Inclusion Law, which makes accessibility mandatory on websites maintained by government bodies [BRASIL 2015], and reinforced by the Digital

Government Law [BRASIL 2021]. At the technical level, the guidelines are set out in the Electronic Government Accessibility Model (e-MAG) and, more recently, in the ABNT NBR 17225 standard, which establishes accessibility requirements for web content and applications [ABNT 2025], both aligned with the WCAG [W3C 2018]. The relevance of the topic is also quantitative: in 2022, Brazil had 18.6 million people with disabilities, equivalent to 8.9% of the population aged two years or older [IBGE 2023].

In the municipality, citizens interact with the government through different digital channels. The city hall portal concentrates the services and information of the Executive Branch, and the city council portal provides access to the legislative activity and to the transparency of the Legislative Branch. The ombudsman channel, in turn, receives users' submissions, such as complaints, reports, suggestions and requests, in accordance with Law No. 13.460/2017 [BRASIL 2017]. The ombudsman office is, therefore, the means by which citizens can point out failures in the public service itself, including the lack of accessibility. If the channel is not accessible, the person with a disability is prevented even from registering this complaint. In Mato Grosso, a previous study had already identified limitations in access to and in the provision of information by municipal ombudsman offices [Zahn et al. 2023].

The accessibility of government portals has been evaluated mainly at the federal level, with successive studies on the portals of the ministries and of the Gov.br platform [Oliveira and Eler 2015, Oliveira et al. 2020, Sousa et al. 2024, Barros et al. 2024]. At the municipal level, Brazilian studies are still based on small samples, such as the city halls of the municipalities that host the campuses of a university [Dorneles and Melo 2024] or 13 portals from a region of Ceará [de França et al. 2025]. In the international scenario, there are larger-scale evaluations, such as that of 7,713 home pages of Italian municipalities [Valtolina and Fratus 2022] and that of 342 U.S. counties, which also investigated factors associated with accessibility, such as financial resources and population density [Bai et al. 2021]. Recent studies also associate the level of accessibility with organizational factors, such as oversight by other spheres of government and the periodic review of websites [Frاندell and Allgood 2026]. In general, however, these evaluations consider only the home page of the portal, including in recent international comparisons [Santos et al. 2026]. This limitation is especially relevant for municipal channels, since the ombudsman channel and the city council portal are often hosted on another domain or on vendor systems. It is therefore not known whether the score of the city hall portal represents that of the other channels of the same municipality.

Mato Grosso offers a suitable context for this investigation. The state has 142 municipalities distributed over about 903 thousand km², with a population density of approximately 4 inhabitants per km² [IBGE 2026]. In this extensive and sparsely populated territory, digital channels can bring citizens and government closer together, reducing the need for physical travel. In addition, the state instituted the Digital Strategic Agenda 2023–2027, which has as one of its goals “to adapt all digital services of the state Executive Branch to the e-MAG accessibility standard, by 2026” [MATO GROSSO 2023]. Although the goal applies to the state Executive Branch, and not to the municipalities, it signals the priority of the topic in the state, and a municipal diagnosis can serve as a baseline for monitoring digital accessibility in Mato Grosso.

In view of this, this work evaluates the digital accessibility of the city hall portals, of the city council portals and of the electronic ombudsman channels of the 142 municipalities of Mato Grosso, with two automated tools based on WCAG 2.1. The study seeks to answer the following research questions:

- **RQ1:** What is the level of accessibility of the city hall portals, of the city council portals and of the ombudsman channels?
- **RQ2:** Does the accessibility score of one environment represent that of the other environments of the same municipality?
- **RQ3:** Is the type of ombudsman solution associated with the difference between the score of the ombudsman channel and that of the city hall portal?
- **RQ4:** Is the level of accessibility associated with the population size and with socioeconomic indicators of the municipalities?

In summary, the *Research Context* of this work is digital accessibility as a legal and social requirement of municipal public services. The *Scientific and/or Practical Problem* is the absence of systematic evidence on the accessibility of the portals of the city halls and city councils of Mato Grosso and on its maintenance in the electronic ombudsman channels. As the *Proposed Solution and/or Analysis*, the three environments are evaluated, in a study that covers all the municipalities of the state. The *Related IS Theory* adopted is the DeLone and McLean IS Success Model [DeLone and McLean 1992, DeLone and McLean 2003], in which system quality, here operationalized by compliance with accessibility guidelines, is an antecedent of use and user satisfaction. The *Research Method* is described in Section 3, the *Summary of Results* is presented in Section 4, and the *Contributions and Impact to IS area* are discussed in Section 5. Section 2 presents the related work.

2. Related work

This section presents the IS theory that underpins the study, the works that evaluated the accessibility of government portals, the studies on participation channels and factors associated with accessibility and, finally, the positioning of this work in relation to the literature.

2.1. Related IS Theory

The DeLone and McLean Information Systems Success Model [DeLone and McLean 1992, DeLone and McLean 2003] explains the success of an IS through interrelated dimensions: system quality, information quality and service quality influence intention to use, use and user satisfaction, which in turn determine net benefits. In this work, digital accessibility is treated as a component of system quality. A portal or ombudsman channel that does not meet accessibility guidelines restricts use by people with disabilities and, with it, the benefits that the digital public service can generate. The automated evaluation therefore measures an antecedent of effective use and of citizen satisfaction, and not use itself.

2.2. Accessibility in government portals

At the federal level, the accessibility of Brazilian portals has been monitored by a sequence of studies. The evaluation of 39 ministry websites, with code validators, showed

that the portals still did not adequately implement the Electronic Government standards, with errors that compromised access to information [Oliveira and Eler 2015]. The re-evaluation of these portals in 2020 indicated advances in compliance with e-MAG and reinforced the importance of periodic evaluations [Oliveira et al. 2020]. A later replication study evaluated 36 federal portals, of which 17 allowed an analysis of evolution, and observed that e-MAG, from 2014, is outdated in relation to the most recent version of the WCAG [Sousa et al. 2024]. The Gov.br platform was also evaluated with three automated tools (ASES, AccessMonitor and TAW) and a contrast checker, and the results indicated that the portal did not meet the minimum accessibility requirements [Barros et al. 2024]. At the state level, the evaluation of the websites of the state governments with two automated tools found barriers in most states and discussed the results in light of social and economic indicators [Carvalho et al. 2017].

At the municipal level, Brazilian studies are still few and small in scale. The home pages of the city halls of the municipalities that host the Unipampa campuses were evaluated with ASES, and most presented accessibility errors [Dorneles and Melo 2024]. In another study, 13 city hall portals from the Sertões de Crateús region, in Ceará, served as a case to compare the evaluation made by a language model with that of three automated tools [de França et al. 2025]. With a focus on transparency, and not on accessibility, a standardized methodology, based on the ISO/IEC 25000 series, was proposed to evaluate municipal transparency portals [Amorim and de Menezes 2016]. In the international scenario, the evaluation of 7,713 home pages of Italian municipalities with AChecker and VaMolà revealed a low level of accessibility [Valtolina and Fratus 2022], and the study of 342 U.S. county portals found a positive relationship between accessibility and usability [Bai 2019]. In common, these works evaluate only the home page of the main portal. Even in a recent international comparison, the analysis was restricted to home pages [Santos et al. 2026]. No studies were found that evaluate the portals of city councils.

2.3. Participation channels and factors associated with accessibility

Citizen participation channels have received less attention. In Mato Grosso, the documentary analysis of 20 municipal ombudsman offices showed that only 15% met all the requirements of access and of provision of information to citizens [Zahn et al. 2023]. In a proposal for citizen participation through wearable devices, it was observed that the adoption of these channels depends not only on the quality of the technology, but also on the citizen's trust in the use that public management makes of the information [Giangareli et al. 2026].

Other studies investigate what explains the level of accessibility. In U.S. counties, population density and website complexity were the main predictors, while the budget had only a marginal effect [Bai et al. 2021]. The combination of a survey of 621 public managers and the evaluation of the websites showed that oversight by other spheres of government and periodic review favor accessibility [Frاندell and Allgood 2026]. In Brazil, the evaluation of 80 priority public services, which combined AccessMonitor with requests under the Access to Information Law, revealed that agencies usually outsource the responsibility for accessibility testing and lack specific training [da Costa Nunes et al. 2025]. These findings suggest that accessibility may depend both on characteristics of the municipality and on contracting and management decisions, which motivates questions RQ3

and RQ4.

2.4. Positioning of this work

Table 1 compares the main related works with this study.

Table 1. Comparison between the related works and this study

Work	Sphere	Sample	Instruments	Environments
[Oliveira and Eler 2015]	Federal	39 ministries	Code validators	Home page
[Sousa et al. 2024]	Federal	36 portals	ASES, Wappalyzer and manual inspection	Home page
[Barros et al. 2024]	Federal	Gov.br portal	ASES, AccessMonitor and TAW	Gov.br portal
[Carvalho et al. 2017]	State	State governments	Two tools and one metric	Home page
[Dorneles and Melo 2024]	Municipal	City halls of the campuses	ASES	Home page
[de França et al. 2025]	Municipal	13 city halls	LLM, IBM, AChecker and WAVE	Home page
[Valtolina and Fratus 2022]	Municipal (Italy)	7,713 municipalities	AChecker and VaMolà	Home page
[Zahn et al. 2023]	Municipal (MT)	20 ombudsman offices	Documentary analysis	Ombudsman
This work	Municipal (MT)	142 municipalities	AMASWeb and AccessMonitor	City hall, city council and ombudsman

The related works show that, despite the legislation and the standards, accessibility failures persist in all spheres of government. This work differs in three aspects. First, it covers all 142 municipalities of Mato Grosso, while Brazilian municipal studies were limited to a few city halls. Second, it evaluates three environments of the same municipality (the city hall portal, the city council portal and the ombudsman channel), which makes it possible to verify whether the score of one environment represents that of the others. Third, it uses two automated tools, which reduces the dependence on the criteria of a single tool. The study also provides a baseline for monitoring municipal digital accessibility in the state.

3. Methodology

As for the *Research Method*, this is a descriptive, comparative study with a quantitative approach, of a census nature, covering the 142 municipalities of Mato Grosso. For each municipality, three environments were considered: the institutional city hall portal, the city council portal and the electronic ombudsman channel. Each environment was evaluated with two automated tools based on WCAG 2.1 [W3C 2018], AMASWeb and AccessMonitor, and the scores were compared between the environments of the same municipality and related to municipal indicators. The descriptive statistics of the scores answer RQ1; the paired comparison between the environments, RQ2; the comparison by type of ombudsman solution, RQ3; and the correlation with the municipal indicators, RQ4. Figure 1 summarizes the process, organized into three phases: survey, automated evaluation, and consolidation and analysis.

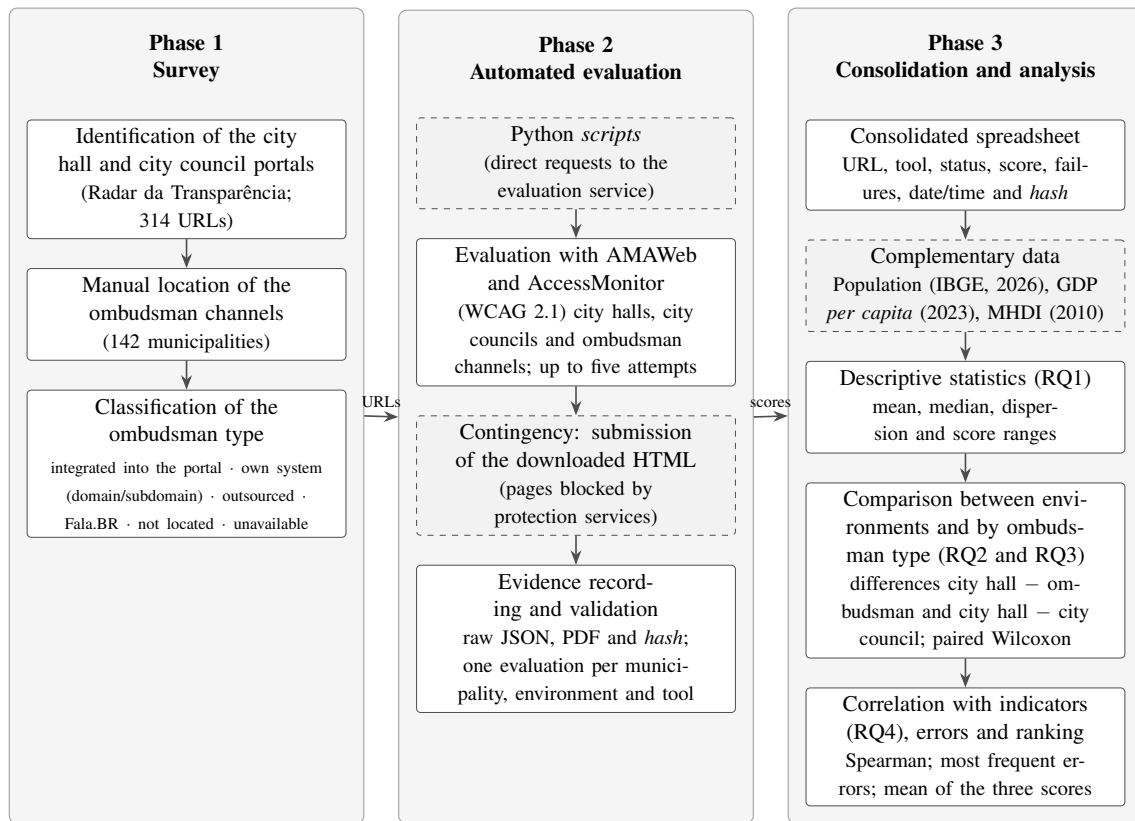


Figure 1. Methodological process adopted in the evaluation of the accessibility of the city hall portals, the city council portals and the ombudsman channels of the municipalities of Mato Grosso.

3.1. Population and source of the addresses

The addresses of the portals were obtained from the Radar da Transparência Pública [ATRICON 2025], a dashboard of the National Public Transparency Program that gathers the evaluated public entities and their web addresses. The entities linked to the municipalities of Mato Grosso in the 2025 cycle were extracted, and each address was classified as city hall, city council or other entity, based on the domain (for example, *.leg.br* for city councils) and on the page title. The portals of the city halls and of the city councils were analyzed. The location of the ombudsman channels was carried out on September 21, 2026, and the evaluations were carried out between September 20 and 22, 2026.

3.2. Identification and classification of the ombudsman channels

The ombudsman channels were located manually, by navigation from the official portal of each municipality. For each municipality, the following were recorded in a spreadsheet:

- the official portal and the starting URL;
- the final URL reached, or the last one attempted;
- the page title;
- the navigation path followed (for example, *Portal → Ombudsman → New protocol*);
- a description of the evidence found;

- the status of the channel: confirmed form, channel without form, intermediate portal, page without visible content, access error or not verifiable.

The URL submitted to the tools was the final URL, that is, the page on which the citizen registers the submission. For the municipalities whose ombudsman channel was left without a valid evaluation, the status was defined from the failure records of the collection and, when these records were not sufficient, by a new manual verification, carried out on September 29, 2026.

Each ombudsman channel was classified into one of the following types, based on the hosting address and on the recorded status: (i) integrated into the portal, when it is on the same domain as the city hall portal; (ii) own system, when it is on a domain or subdomain distinct from that of the city hall; (iii) outsourced, when it is hosted on a vendor's domain; (iv) Fala.BR, the platform of the Office of the Comptroller General (Controladoria-Geral da União); (v) not located; and (vi) unavailable at the time of collection. The classification considers only the hosting address. For this reason, a system classified as own may be the same vendor solution replicated under the domain of several municipalities, and simple forms were not distinguished as a separate type.

3.3. Evaluation instruments

Two automated accessibility evaluation tools based on WCAG 2.1 were used. The use of more than one tool, also adopted in previous studies [Barros et al. 2024], reduces the dependence on the criteria and limitations of a single evaluator.

AMAWeb [AMAWEB 2026], developed by the Federal University of São Paulo (UNIFESP) and by the Federal Institute of Rio Grande do Sul (IFRS), evaluates a page from its URL or from its HTML code. It assigns a score from 1 to 10 and classifies each verified practice as acceptable, not acceptable or to be verified manually.

AccessMonitor [AMA 2026], the validator of the Portuguese Web Accessibility Observatory, maintained by the Agency for Administrative Modernization (AMA) of Portugal, runs the QualWeb engine [QualWeb 2026]. QualWeb applies ACT rules, WCAG 2.1 techniques and best practices, and on these results the tool calculates a score from 1 to 10. Because it is open source, the tool was run on a local instance, installed from the official repositories, in the following versions: `accessmonitor-docker` service (June 2026), `QualWeb core` 0.8.11, `act-rules` 0.8.0, `wcag-techniques` 0.4.7, `best-practices` 0.7.12 and the `accessmonitor-rulesets` 1.0.4 rule set.

Both tools return reports with the same structure: score, practices classified by the three verdicts, count of HTML elements and page title. Since the scales of the tools are not equivalent, the descriptive statistics, the comparison between environments and the error analysis were carried out separately for each tool.

3.4. Collection procedure

The collection was automated by Python *scripts*, one for each tool:

- **AMAWeb:** the *script* queries the HTTP service used by the evaluator's own web interface, which receives the encoded URL and returns the report in JSON. This service has no formal public documentation.

- **AccessMonitor:** the *script* queries the local instance, which receives the Base64-encoded URL or the HTML code of the page. The requests indicate the authorized origin in the `Referer` header, as required by the instance.

For each URL, the *scripts* preserved the JSON returned by the tool, without changes, and generated a PDF report. The requests were executed with three simultaneous evaluations and a time limit per evaluation. Each run saved its results in a directory identified by the start date and time (YYYY-MM-DD/HH-MM-SS), accompanied by a CSV spreadsheet with the result of each URL and a manifest with the parameters used. URLs without a valid result were resubmitted in successive rounds, up to the limit of five attempts.

Pages that blocked direct access by the tool were evaluated by an alternative mode: the HTML of the page was downloaded with the headers of a common browser and submitted to the HTML code evaluation mode. In this mode, the page's *scripts* are not executed, so the submission mode (URL or HTML) was recorded in each evaluation. Websites with an invalid TLS certificate were not evaluated, since certificate validation was not bypassed.

3.5. Validity criteria

An evaluation was considered valid when the analyzed document corresponds to the requested page. Evaluations of the following were discarded: (i) challenge or access-denied pages of protection services, identified by markers in the title ("*Just a moment*", "*Attention Required!*", "*Access denied*"); (ii) error pages, with an HTTP code equal to or greater than 400 or whose title indicates an error (for example, "*404 - File or directory not found*"); and (iii) documents with fewer than ten HTML elements. Only the ombudsman channels with a confirmed submission form are included in the score analysis. Intermediate portals, channels without a form and unavailable channels were excluded, since the corresponding page is not the one on which the citizen registers the submission.

Each municipality contributes a single evaluation per environment and per tool. When there was more than one valid evaluation, priority was given, in this order, to the evaluation with effectively evaluated content, to that of the portal's home page, to that made by URL over that made by HTML and, finally, to the most recent one.

3.6. Data analysis

The reports were consolidated into a spreadsheet with tabs for city councils, city halls and ombudsman channels, in addition to analysis and error tabs. For each evaluation, the following were recorded: URL, tool, page status, score, date and time, number of practices with failure, warning and success, and the SHA-256 *hash* of the source file, which makes it possible to verify the correspondence with the raw report.

The unit of analysis is the municipality. For each tool, each municipality has up to three scores: that of the city hall portal, that of the city council portal and that of the ombudsman channel. The analyses follow two inclusion rules. In the analyses by tool, each environment with a valid evaluation in the tool considered is included, and the comparisons between environments use only the municipalities with both environments evaluated in the same tool. In the analyses with the consolidated score, only the environment evaluated by both tools is included. The scores were described by mean, median, standard

deviation, minimum and maximum, for each environment and tool, and distributed into score ranges (below 5.0; from 5.0 to 6.9; from 7.0 to 8.9; and 9.0 or more). The ranges are only descriptive, since neither tool defines a score from which a page is considered accessible.

For RQ2, the city hall scores were compared with those of the ombudsman channel and the city hall scores with those of the city council, considering only the municipalities with both environments evaluated in the same tool. The mean difference and the mean absolute difference between the scores were calculated, and the significance of the differences was verified by the paired Wilcoxon test. The difference between city hall and ombudsman channel reflects the path of the citizen, who normally reaches the ombudsman channel from the portal. For RQ3, the scores of the ombudsman channels and the differences in relation to the city hall portal were compared across the solution types.

For RQ4 and for the ranking, the consolidated score of each environment was used, that is, the mean of the scores of the two tools, calculated only when the environment had a valid evaluation in both. Since all the municipalities considered in these analyses were evaluated by both tools, the difference in scale between them affects all of them in the same way. The robustness of this choice was verified by the agreement between the tools (Spearman correlation between the scores of each page) and by comparing the consolidated ranking with the rankings obtained by tool and with standardized scores (z-score). The consolidated scores were related to the municipal indicators through Spearman's correlation coefficient and through means by population range, the latter calculated over the same municipalities as the ranking. The indicators used were the population estimated by IBGE (2026), the GDP *per capita* (IBGE, 2023) and the MHDI (2010), obtained from the Atlas of Human Development in Brazil. In all tests, a significance level of 0.05 was adopted. The analyses were carried out in Python, with the pandas and SciPy libraries.

The most frequent errors were determined by the proportion of pages with at least one occurrence of each practice classified as not acceptable. Each practice was associated with the corresponding WCAG 2.1 success criterion.

Finally, the municipalities were ranked by the simple mean of the consolidated scores of the three environments, considering only the municipalities with the three environments evaluated by both tools. Ties were broken by alphabetical order, and the ten best-ranked and the ten worst-ranked municipalities are presented.

The raw reports, the consolidated spreadsheet and the collection *scripts* are available at XX.

4. Results and discussion

4.1. Characterization of the collection

From the Radar da Transparência, 314 unique URLs were extracted, among portals of city halls, of city councils and of other entities, such as intermunicipal consortia. In locating the ombudsman channels, the submission form was confirmed in 127 of the 142 municipalities (89.4%). In the others, the following were found: intermediate portals, which led to a home page or to a service catalog, and not to the form (11), channels without an ombudsman form, such as contact pages (3), and one channel unavailable

due to a DNS failure. Since some municipalities share the same platform, the 142 records correspond to 136 unique URLs. Seven municipalities (Araguainha, Araputanga, Barra do Garças, Cláudia, Paranaíta, Sorriso and Várzea Grande) use Fala.BR, and the evaluation considered the same general registration page for all of them; in the case of Araputanga, the municipality's specific address returned an error. Two other municipalities (Canarana and Cáceres) direct the citizen to this platform.

Table 2 summarizes the coverage of the collection by environment and by tool. For the ombudsman channel, the table considers only the confirmed forms. With at least one tool, the three environments were evaluated in 88 municipalities, and the city hall and the ombudsman channel, in 110. With both tools, the three environments were evaluated in 75 municipalities, which make up the basis of the ranking. The study therefore covered all the municipalities of the state, but the complete analysis of the three environments by both tools gathers just over half of them.

Table 2. Municipalities with a valid evaluation, by environment and tool

Environment	At least one	AMASWeb	AccessMonitor	Both
City hall	123	117	123	117
City council	107	100	107	100
Ombudsman (confirmed form)	117	101	116	100
The three environments	88	75	88	75

Total municipalities: 142. *At least one*: the environment has a valid evaluation in at least one tool, which may differ between the environments of the same municipality. *Both*: the environment has a valid evaluation in both tools.

The coverage of AMASWeb was lower because of persistent failures: 47 of the 314 Radar addresses remained without a response after five attempts. These failures did not result from request rate limiting, since, in the same period, URLs already evaluated were resubmitted and processed normally, while the addresses with failures repeated the error also in the evaluator's web interface, including when submitted through the HTML code. Blocking by protection services was the main cause of data loss. In AccessMonitor, 51 unique URLs returned a Cloudflare block page (33 portals and 18 ombudsman channels). There were also 33 internal errors of the service when processing the QualWeb report, 18 timeouts in the download of the HTML, 5 DNS failures and 3 invalid TLS certificates. The ombudsman channels hosted by the same vendor failed in the same way in both tools, which shows that the adopted platform affects even the possibility of evaluating the channel. The contingency through the downloaded HTML was decisive for the coverage of AccessMonitor, with 128 valid evaluations obtained in this mode. The means obtained in the two submission modes were close (in AMASWeb, 7.73 by URL and 7.64 by HTML; in AccessMonitor, 7.15 and 6.89), and therefore the evaluations of the two modes were analyzed together.

4.2. Level of accessibility of the environments (RQ1)

Figure 2 presents the distribution of the scores by environment and tool, and Table 3, the descriptive statistics. The ombudsman channels obtained the highest means and the lowest dispersion in both tools. The city hall portals presented the greatest dispersion, with the largest standard deviations, and the city council portals concentrated the lowest

outliers. In all environments, AMAWeb assigned higher scores than AccessMonitor: in the 317 municipality–environment pairs evaluated by both tools, the mean difference was 0.67 points.

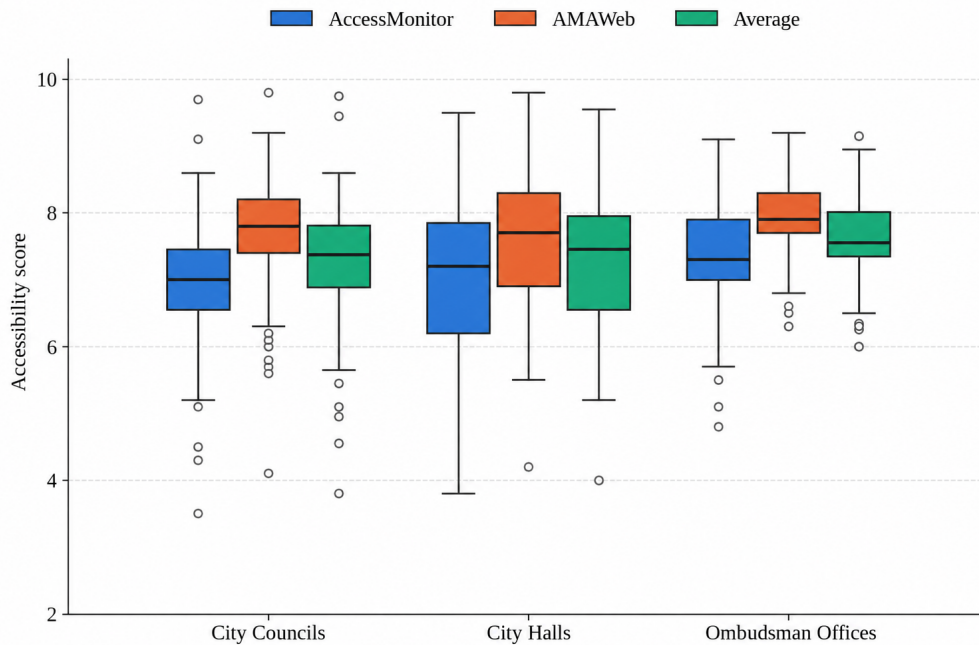


Figure 2. Distribution of the accessibility scores by environment and tool. The Mean series corresponds to the mean of the scores of the two tools.

Most pages fell in the range of 7.0 to 8.9 points in all environments. The proportion of pages with a score equal to or greater than 9.0 was small, especially in the city councils (3.0% in AMAWeb and 1.9% in AccessMonitor), while scores below 7.0 were more frequent in the city hall and city council portals evaluated by AccessMonitor (39.8% and 46.8%, respectively).

Table 4 lists the practices most frequently classified as not acceptable in AMAWeb. Insufficient color contrast and links without an accessible name were the most common problems in the three environments. The error profiles, however, differ. In the city hall portals, markup problems stand out, such as repeated `id` attributes (65.0%). In the city council portals, links without an accessible name (82.0%) and the absence or duplication of the `h1` heading (74.0%) stand out. In the ombudsman channels, failures associated with keyboard navigation predominate, such as the absence of a mechanism to skip to the main content and the removal of focus by JavaScript (75.2% each). These failures are indications of barriers for those who use a keyboard or screen reader when filling out a form; confirmation of the actual impact requires manual inspection and user testing. The main language marked incorrectly appears in 55.4% of the ombudsman channels, against 5.1% of the city hall portals and 3.0% of the city council portals, which is compatible with the replication of the same platform in several municipalities. AccessMonitor reproduced the general pattern: insufficient contrast in 74.8% of the city hall portals, 72.9% of the city council portals and 68.1% of the ombudsman channels, and absence of a link to skip to the content in 69.0% of the ombudsman channels.

Table 3. Descriptive statistics of the accessibility scores, by environment and tool

Measure	City hall		City council		Ombudsman	
	AMASWeb	AccessM.	AMASWeb	AccessM.	AMASWeb	AccessM.
<i>n</i>	117	123	100	107	101	116
Mean	7.65	7.01	7.68	6.89	7.98	7.48
Median	7.70	7.20	7.80	7.00	7.90	7.30
Standard deviation	1.00	1.22	0.88	1.15	0.68	0.91
Minimum	4.20	3.80	4.10	3.50	6.30	4.80
Maximum	9.80	9.50	9.80	9.70	9.20	9.10
<i>Distribution by score range (%)</i>						
Below 5.0	0.9	6.5	1.0	7.5	0.0	0.9
5.0 to 6.9	24.8	33.3	14.0	39.3	5.0	12.1
7.0 to 8.9	65.0	54.5	82.0	51.4	79.2	75.0
9.0 or more	9.4	5.7	3.0	1.9	15.8	12.1

Table 4. Practices most frequently classified as not acceptable in AMASWeb

Error	Criterion WCAG	Level	Pages with the error (%)		
			City halls (<i>n</i> = 117)	City councils (<i>n</i> = 100)	Ombudsman (<i>n</i> = 101)
Insufficient color contrast	1.4.3	AA	84.6	78.0	73.3
Links without an accessible name	4.1.2	A	76.1	82.0	58.4
Repeated id attributes	4.1.1	A	65.0	45.0	15.8
Image links without alternative text	2.4.4	A	59.0	58.0	56.4
Missing h1 or duplicated h1	1.3.1	A	51.3	74.0	57.4
No link to skip to the content	2.4.1	A	46.2	45.0	75.2
Focus removed by JavaScript	2.4.7	AA	41.0	16.0	75.2
Heading hierarchy violated	2.4.10	AAA	40.2	37.0	67.3
Main language marked incorrectly	3.1.1	A	5.1	3.0	55.4

4.3. Consistency between the environments of the same municipality (RQ2)

Table 5 compares the city hall scores with those of the ombudsman channel and of the city council, in the municipalities with both environments evaluated in the same tool. The ombudsman channel obtained a higher score than the city hall portal in 53 of the 96 municipalities in AMASWeb and in 50 of the 109 in AccessMonitor, with mean differences of -0.38 and -0.43 points. The difference was statistically significant in AMASWeb, but not in AccessMonitor. In the comparison with the city council, the means were practically equal, and the differences were not significant in either tool.

The mean absolute difference, however, was close to one point in both comparisons and in both tools. This means that, even when the overall means are similar, the scores of the same municipality vary relevantly between the environments, sometimes in favor of one, sometimes of the other. The score of the city hall portal, therefore, does not represent that of the ombudsman channel nor that of the city council portal, and each environment needs to be evaluated separately.

Table 5. Paired comparison between the city hall portal and the other environments of the same municipality

Measure	City hall × Ombudsman		City hall × City council	
	AMASWeb	AccessM.	AMASWeb	AccessM.
Municipalities compared	96	109	93	99
Other environment with higher score	53	50	45	43
Other environment with lower score	38	52	45	52
Equal scores	5	7	3	4
Mean difference (city hall – other)	−0.38	−0.43	0.00	0.17
Mean absolute difference	0.92	1.17	0.98	1.19
Wilcoxon test (<i>p</i>)	0.007	0.094	0.937	0.287

4.4. Types of ombudsman solution (RQ3)

Table 6 presents the frequency of each type of ombudsman solution, the mean scores of the confirmed forms and the difference in relation to the city hall portal.

Table 6. Types of ombudsman solution, mean scores of the ombudsman channel and mean difference city hall – ombudsman channel, by tool

Type	Municipalities		Ombudsman score (<i>n</i>)		Mean difference (pairs)	
			AMASWeb	AccessM.	AMASWeb	AccessM.
Integrated into the portal	37	26.1	7.63 (15)	7.11 (28)	−0.09 (13)	0.08 (23)
Own system	59	41.5	7.83 (55)	7.24 (57)	−0.02 (53)	0.17 (56)
Outsourced	36	25.4	8.54 (24)	8.25 (24)	−1.30 (23)	−2.07 (23)
Fala.BR	9	6.3	8.00 (7)	8.30 (7)	−0.57 (7)	−1.46 (7)
Unavailable	1	0.7	—	—	—	—
Total	142	100				

Scores calculated only on the confirmed forms with a valid evaluation in each tool. A negative difference indicates an ombudsman channel with a higher score than the city hall portal.

Own systems on a distinct domain or subdomain are the most frequent type (41.5%), and 54 of the 59 follow the same address pattern (`ouvidoria.<municipality>.mt.gov.br/Manifestacao/`). This suggests that part of the channels classified as own is, in fact, the same solution replicated across municipalities, possibly provided by a single company and hosted under the city hall's domain. The same occurs among the outsourced ones, which are concentrated in a few vendors. Those integrated into the portal account for 26.1%, the outsourced ones for 25.4%, and 9 municipalities (6.3%) direct the citizen to Fala.BR.

In the ombudsman channels integrated into the portal and in the own systems, the mean difference in relation to the city hall portal is close to zero. The outsourced ombudsman channels, on the other hand, obtained higher scores than the respective portal in 18 of the 23 municipalities compared, in both tools, with mean differences of 1.30 and 2.07 points. The data show an association between the type of ombudsman channel and the score difference, but do not allow it to be attributed to the vendor or to the municipality's accessibility policy. In the case of Fala.BR, the evaluated municipalities use the same registration page; therefore, their scores do not represent independent implementations.

4.5. Relationship with population size and socioeconomic indicators (RQ4)

Before the analyses with the consolidated score, the agreement between the tools was verified. The correlation between the AMAWeb and AccessMonitor scores for the same page was 0.91 in the city hall portals, 0.79 in the city council portals and 0.71 in the ombudsman channels, and the difference between the tools was approximately constant (between 0.49 and 0.85 points, depending on the environment). Under these conditions, the mean of the two scores largely preserves the order of the municipalities, which was confirmed in the ranking (Section 4.6).

Table 7 presents the Spearman correlation coefficients between the consolidated scores and the municipal indicators. The only significant association was the weak negative correlation between GDP *per capita* and the score of the city hall portals ($\rho = -0.268$; $p = 0.004$). In the ombudsman channels, the correlation with population was positive and weak, but not significant ($\rho = 0.194$; $p = 0.053$). In the city council portals, no association was significant.

Table 7. Spearman correlation (ρ) between the consolidated scores and the municipal indicators

Indicator	City hall	City council	Ombudsman
Population (IBGE, 2026)	-0.155	0.029	0.194
GDP <i>per capita</i> (2023)	-0.268**	-0.038	-0.041
MHDI (2010)	-0.171	-0.042	-0.025
<i>n</i>	117	100	100

* $p < 0.05$; ** $p < 0.01$.

Table 8 presents the means by population range in the 75 municipalities of the ranking base. The scores did not increase consistently with population size. In the ombudsman channels, the highest mean occurred among the municipalities with more than 50 thousand inhabitants, but this stratum contains only eight observations.

Table 8. Mean consolidated scores by population range

Population range	<i>n</i>	City hall	City council	Ombudsman	Mean
Up to 10 thousand inhabitants	27	7.57	7.14	7.76	7.49
From 10 to 50 thousand inhabitants	40	7.09	7.36	7.78	7.41
More than 50 thousand inhabitants	8	7.22	7.31	8.21	7.58

4.6. Ranking of the municipalities

The ranking was calculated for the 75 municipalities with the three environments evaluated by both tools. For each environment, the consolidated score was calculated, and the final score corresponds to the simple mean of the three consolidated scores. In case of a tie, alphabetical order was adopted. The range corresponds to the difference between the highest and the lowest consolidated score of the municipality. Tables 9 and 10 present the ten best-ranked and the ten worst-ranked municipalities; the complete list is available at XX.

Table 9. Ten municipalities with the highest consolidated accessibility means
($n = 75$)

#	Municipality	City hall	City council	Ombudsman	Mean	Range
1	Nossa Senhora do Livramento	8.95	9.45	7.35	8.58	2.10
2	Conquista d'Oeste	9.30	7.50	8.65	8.48	1.80
3	Sinop	9.55	8.30	7.50	8.45	2.05
4	Cuiabá	8.00	9.75	7.00	8.25	2.75
5	Porto Esperidião	8.75	8.60	7.35	8.23	1.40
6	Vila Rica	8.40	8.45	7.55	8.13	0.90
7	Vila Bela da Santíssima Trindade	7.90	8.50	7.90	8.10	0.60
8	Rondolândia	7.80	7.75	8.65	8.07	0.90
9	Itanhangá	6.80	8.20	9.15	8.05	2.35
10	Tapurah	7.55	7.25	9.15	7.98	1.90

Table 10. Ten municipalities with the lowest consolidated accessibility means
($n = 75$)

#	Municipality	City hall	City council	Ombudsman	Mean	Range
66	Marcelândia	5.50	7.75	7.40	6.88	2.25
67	Santa Rita do Trivelato	5.40	6.40	8.85	6.88	3.45
68	Matupá	6.55	7.70	6.35	6.87	1.35
69	Pedra Preta	5.75	7.05	7.55	6.78	1.80
70	São Pedro da Cipa	8.30	5.45	6.30	6.68	2.85
71	Nova Marilândia	7.90	4.55	7.40	6.62	3.35
72	Nortelândia	7.50	4.95	7.35	6.60	2.55
73	Querência	5.80	5.95	7.35	6.37	1.55
74	Itiquira	6.45	6.55	6.00	6.33	0.55
75	Arenápolis	6.05	3.80	6.95	5.60	3.15

The consolidated ranking proved to be robust. Its correlation with the ranking made only with AccessMonitor was 0.95, and with the one made only with AMAWeb, 0.87; with scores standardized by z-score, it was 0.97. Eight of the top ten coincide with those of the rankings by tool, and nine with those of the standardized ranking. The ranking also reinforces the variation between the environments of the same municipality: the mean range was 1.57 points, and 25 of the 75 municipalities had a difference of 2 points or more between the best and the worst environment. Even among the best-ranked there are municipalities with one environment well below the others, such as Cuiabá, whose ombudsman channel obtained 7.00, while the city council portal obtained 9.75.

4.7. Discussion

In light of the DeLone and McLean IS Success Model [DeLone and McLean 1992, DeLone and McLean 2003], the results indicate that system quality, operationalized by compliance with accessibility guidelines, is not homogeneous across the channels of the same municipality. Since the ombudsman channel and the city council portal are usually accessed from the city hall portal, failures in any of these environments can compromise the path of the citizen and, with it, the use of and satisfaction with the service. This finding indicates, for the municipal context, the limit of evaluations restricted to the home page [Valtolina and Fratus 2022, Dorneles and Melo 2024, de França et al. 2025].

Insufficient contrast and the lack of accessible names in links, present in the three environments, are failures that are relatively simple to correct, which depend more on design guidelines than on structural changes in the systems. The concentration of failures associated with keyboard navigation in the ombudsman channels deserves attention, since these channels exist precisely to receive submissions, including about the lack of accessibility [BRASIL 2017]. The location of the channels also revealed that 14 municipalities do not lead the citizen directly to a form, but to intermediate portals or to contact pages.

The weak correlations with the socioeconomic indicators are compatible with the study of U.S. counties, in which the budget had only a marginal effect on accessibility [Bai et al. 2021]. The negative correlation between GDP *per capita* and the score of the city hall portals, although weak, indicates that greater economic capacity does not translate, by itself, into more accessible portals. The performance of the outsourced ombudsman channels and the replication of the same solution in dozens of municipalities raise the hypothesis that accessibility depends more on the contracted solutions and their management than on the size of the municipality. This hypothesis is consistent with the findings on the outsourcing of responsibility for accessibility testing [da Costa Nunes et al. 2025] and on the role of oversight and periodic review of websites [Frاندell and Allgood 2026], but contracting and management decisions were not measured in this study and should be investigated directly in future work.

5. Final remarks

This study covered the 142 municipalities of Mato Grosso and evaluated the digital accessibility of the city hall portals, of the city council portals and of the electronic ombudsman channels, with two automated tools based on WCAG 2.1, AMAWeb and AccessMonitor. The ombudsman form was confirmed in 127 municipalities, and the three environments had a valid evaluation in 88 municipalities with at least one tool and in 75 with both.

Regarding RQ1, the ombudsman channels obtained the highest means and the lowest dispersion, and most pages of the three environments were between 7.0 and 8.9 points. Insufficient contrast and links without an accessible name were the most common problems, and the ombudsman channels concentrated failures associated with keyboard navigation and focus, which are indications of barriers precisely for those who use a keyboard or screen reader when registering a submission. Regarding RQ2, the score of the city hall portal did not represent that of the other environments: the mean absolute difference between environments of the same municipality was close to one point, both in relation to the ombudsman channel and to the city council. Regarding RQ3, the outsourced ombudsman channels obtained higher scores than those of the respective portals in 18 of the 23 municipalities compared, and the replication of the same solution in dozens of municipalities indicates that part of the results reflects the adopted platform. Regarding RQ4, the only significant association was the weak negative correlation between GDP *per capita* and the score of the city hall portals, and the data do not support the interpretation that larger municipalities or those with better socioeconomic indicators offer more accessible pages.

The results also show that the existence of an electronic channel does not ensure adequate access conditions. In 14 municipalities, the path found leads to an intermediate portal or to a contact page, and not to a submission form. In addition, blocking by protec-

tion services, internal errors of the tools, timeouts, DNS failures and invalid certificates reduced the coverage of the collection; these occurrences may equally hinder citizens' access to digital public services.

As for the *Contributions and Impact to IS area*, the work offers an unprecedented diagnosis of municipal digital accessibility in Mato Grosso, covering all the municipalities of the state, shows that each channel of the municipality needs to be evaluated separately and provides a replicable collection procedure and a baseline for monitoring digital government in the state. In light of the DeLone and McLean IS Success Model, the results show that system quality, a condition for citizen use and satisfaction, still presents weaknesses in the municipal channels, including those intended for participation. As a practical implication, it is recommended that municipal administrations include accessibility requirements in the processes of contracting and updating portals and ombudsman systems, maintain periodic evaluation routines and require the correction of the failures identified in the shared components.

The study has limitations. The automated scores synthesize only the criteria verifiable by the tools and do not replace manual inspections, tests with assistive technologies or the participation of people with disabilities. A single page of each environment was evaluated, and the research represents a temporal snapshot, since government pages are continuously changed. The use of the downloaded HTML as a contingency does not fully reproduce content generated by *scripts* or dynamic interactions. The status of part of the ombudsman channels without a valid evaluation was defined from the failure records of the collection and from a later manual verification. In addition, the ranking considered only the 75 municipalities with the three environments evaluated by both tools. This decision ensures that all are compared on the same basis, since AMAWeb assigns systematically higher scores than AccessMonitor, but excludes 67 municipalities from the ranking, generally due to failures of one of the tools, to environments not evaluated or to ombudsman channels without a confirmed form. The consolidated score also combines tools with scales that are not equivalent; although the ranking proved robust to the choice of tool and to the standardization of the scores, it should be understood as a diagnostic instrument and a baseline for monitoring, and not as a certification of compliance.

For future work, it is proposed to complement the automated evaluation with manual inspections and tests with users with disabilities, to evaluate other pages of each environment, to directly investigate the role of contracting and management decisions for the digital solutions, to longitudinally monitor the evolution of the municipalities and to extend the analysis to other digital public services. In this way, the diagnosis can support digital government policies oriented not only to the availability of services, but also to their effective, autonomous and equitable use by the entire population.

Declaration of use of artificial intelligence

During the preparation of this work, the authors used the artificial intelligence assistant Claude (Anthropic) to support the preparation of the code of the figure that represents the methodological process (Figure 1), the textual revision of the manuscript and its translation from Portuguese into English. All content generated with the support of the tool was reviewed and validated by the authors, who assume full responsibility for the text, the data and the conclusions presented.

References

- ABNT (2025). ABNT NBR 17225: acessibilidade em conteúdo e aplicações web – requisitos. ABNT, Rio de Janeiro.
- AMA (2026). AccessMonitor: validador automático de acessibilidade web. Agência para a Modernização Administrativa, Portugal. Código-fonte disponível em: <https://github.com/amagovpt/accessmonitor-docker>. Acesso em: 21 set. 2026.
- AMAWEB (2026). AMAWeb: Plataforma de avaliação e monitoramento de acessibilidade na web. Universidade Federal de São Paulo (UNIFESP) e Instituto Federal do Rio Grande do Sul (IFRS). Disponível em: <https://amaweb.unifesp.br/>. Acesso em: 21 set. 2026.
- Amorim, R. and de Menezes, C. (2016). Metodologia de avaliação de portais da transparência municipais. In *Anais do XII Simpósio Brasileiro de Sistemas de Informação*, pages 017–024, Florianópolis, SC, Brasil. SBC.
- ATRICON (2025). Radar da transparência pública. Associação dos Membros dos Tribunais de Contas do Brasil. Disponível em: <https://radardatransparencia.atricon.org.br/>. Acesso em: set. 2026.
- Bai, Y. (2019). The relationship between website accessibility and usability: An examination of U.S. county government online portals. *Electronic Journal of e-Government*, 17(1):47–62.
- Bai, Y., Grzeslo, J., Min, B., and Jayakar, K. (2021). Accessibility of local government websites: influence of financial resources, county characteristics and local demographics. *Universal Access in the Information Society*, 20(4):851–861.
- Barros, Y. S., do Outão, J. C. S., Sacramento, C., Ferreira, S. B. L., Pimentel, M. G., and dos Santos, R. P. (2024). Avaliação de acessibilidade da plataforma gov.br por ferramentas automatizadas. In *Anais do XII Workshop de Computação Aplicada em Governo Eletrônico*, pages 50–61, Brasília, DF, Brasil. SBC.
- BRASIL (2015). Lei n. 13.146, de 6 de julho de 2015: institui a lei brasileira de inclusão da pessoa com deficiência (estatuto da pessoa com deficiência). Diário Oficial da União. Disponível em: https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2015/lei/113146.htm.
- BRASIL (2017). Lei n. 13.460, de 26 de junho de 2017: dispõe sobre participação, proteção e defesa dos direitos do usuário dos serviços públicos da administração pública. Diário Oficial da União. Disponível em: https://www.planalto.gov.br/ccivil_03/_ato2015-2018/2017/lei/113460.htm.
- BRASIL (2021). Lei n. 14.129, de 29 de março de 2021: dispõe sobre princípios, regras e instrumentos para o governo digital e para o aumento da eficiência pública. Diário Oficial da União. Disponível em: https://www.planalto.gov.br/ccivil_03/_ato2019-2022/2021/lei/114129.htm.
- Carvalho, V., Cagnin, M., and Paiva, D. (2017). Avaliação de acessibilidade de web sites de governos estaduais do Brasil. In *Anais do XIII Simpósio Brasileiro de Sistemas de Informação*, pages 116–123, Lavras, MG, Brasil. SBC.

- da Costa Nunes, E. H., Pacheco, S. T., Souza, M. C., and Monteiro, I. T. (2025). Diagnóstico da acessibilidade digital em serviços públicos prioritários usando a lei de acesso à informação. In *Anais do XXIV Simpósio Brasileiro sobre Fatores Humanos em Sistemas Computacionais*, pages 423–445, Belo Horizonte, MG, Brasil. SBC.
- de França, L. C. F., dos S. Andrade, L. M. M., and de Carvalho, W. V. (2025). Um estudo sobre o uso de prompts LLM para a avaliação da acessibilidade web dos portais das prefeituras da região dos sertões dos carateús. In *Anais do XXXI Simpósio Brasileiro de Sistemas Multimídia e Web*, pages 580–584, Rio de Janeiro, RJ, Brasil. SBC.
- DeLone, W. H. and McLean, E. R. (1992). Information systems success: The quest for the dependent variable. *Information Systems Research*, 3(1):60–95.
- DeLone, W. H. and McLean, E. R. (2003). The DeLone and McLean model of information systems success: A ten-year update. *Journal of Management Information Systems*, 19(4):9–30.
- Dorneles, J. and Melo, A. (2024). Avaliação preliminar da acessibilidade de sites de prefeituras dos municípios em que estão situados os campi da unipampa. In *Anais da VIII Escola Regional de Engenharia de Software*, pages 70–78, Santiago, RS, Brasil. SBC.
- Frاندell, A. and Allgood, M. (2026). Breaking barriers: Evaluating accessibility in local government online platforms for individuals with disabilities. *Government Information Quarterly*, 43(1):102111.
- Giangareli, G., Tavares, L., Diniz, W., and Zahn, J. (2026). A wearable architecture for zero-friction civic micro-participation. In *Anais do XIV Latin American Symposium on Digital Government*, pages 13–24, Gramado, RS, Brasil. SBC.
- IBGE (2023). Pesquisa nacional por amostra de domicílios contínua: pessoas com deficiência 2022. Instituto Brasileiro de Geografia e Estatística. Disponível em: https://agenciadenoticias.ibge.gov.br/media/com_mediaibge/arquivos/0a9afaed04d79830f73a16136dba23b9.pdf. Acesso em: 29 set. 2026.
- IBGE (2026). Mato grosso: panorama. Instituto Brasileiro de Geografia e Estatística, Cidades e Estados. Disponível em: <https://www.ibge.gov.br/cidades-e-estados/mt.html>. Acesso em: 29 set. 2026.
- MATO GROSSO (2023). Decreto n. 338, de 20 de junho de 2023: institui a agenda estratégica digital do governo de mato grosso para o período de 2023 a 2027, no âmbito dos órgãos e entidades do poder executivo estadual, e dá outras providências. Diário Oficial do Estado de Mato Grosso. Disponível em: https://iomat.mt.gov.br/legislacao/diario_oficial/detalhes/24062. Acesso em: 18 set. 2026.
- Oliveira, A. and Eler, M. (2015). Acessibilidade em governo eletrônico: um estudo sobre a aplicação de padrões web em sites gov.br. In *Anais do XI Simpósio Brasileiro de Sistemas de Informação*, pages 691–698, Goiânia, GO, Brasil. SBC.
- Oliveira, A., Freire, A., and Eler, M. (2020). Web accessibility evolution in the brazilian government. In *Anais do XVI Simpósio Brasileiro de Sistemas de Informação*, Porto Alegre, RS, Brasil. SBC.

- QualWeb (2026). QualWeb: automated web accessibility evaluator. LASIGE, Faculdade de Ciências da Universidade de Lisboa. Disponível em: <https://github.com/qualweb/qualweb>. Acesso em: 17 set. 2026.
- Santos, C. V., Moreira, E. R., and Santos, I. (2026). Acessibilidade digital em websites governamentais: Um estudo comparativo internacional com base nas diretrizes wcag 2.2. In *Anais do XIV Latin American Symposium on Digital Government*, pages 37–48, Porto Alegre, RS, Brasil. SBC.
- Sousa, F., Moraes, G., Nehme, L., Rivero, L., Freire, A., and Viana, D. (2024). Digital accessibility in brazilian federal government portals: a replication study. In *Anais do XX Simpósio Brasileiro de Sistemas de Informação*, Porto Alegre, RS, Brasil. SBC.
- Valtolina, S. and Fratus, D. (2022). Local government websites accessibility: Evaluation and finding from Italy. *Digital Government: Research and Practice*, 3(3).
- W3C (2018). Web content accessibility guidelines (WCAG) 2.1. W3C Recommendation. Disponível em: <https://www.w3.org/TR/WCAG21/>.
- Zahn, J., Viterbo, J., and Maciel, C. (2023). Tecnologias de informação e comunicação em ouvidorias gerais municipais do estado de mato grosso. In *Anais do XI Workshop de Computação Aplicada em Governo Eletrônico*, pages 165–176, Porto Alegre, RS, Brasil. SBC.